

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd November 2024

SP9 - Enterprise Risk Management

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** You are a CRO of a large-sized capital-management listed company. The company manages capital and investments for retail investors and businesses and invests in multiple risk-weighted instruments to meet the returns for its customers. The company has various functional departments, including marketing, sales, investments, research, operations, etc. All these departments have a head responsible for the respective deliverables, and all heads report directly to the CEO. Additionally, there are some service departments, such as information technology, finance, and human resources, whose heads also report to the CEO. All heads have a dotted reporting line to the CRO, who reports to the CEO.

The company has adopted the approach of acquiring multiple companies. It is moving towards becoming more tech-powered, with its products driven by the latest technology in its design, distribution, and operations. In line with this aspiration, the company has acquired a three-year-old start-up that works in Artificial Intelligence and Machine Learning-based innovative tech solutions.

A week after integrating the company's IT systems with the acquired start-up, a cyber-attack impacted the company's business operations for four days. The incident was reported on social media, and the company's share dipped by 10% the following day after this news.

You are entrusted with handling this cyber crisis and taking immediate steps to curtail and take other necessary steps.

The CEO convened a meeting with the respective heads of the departments and the CRO. In your presentation, apart from aspects related to the cyber incident, you proposed establishing an ERM framework. The CEO responded, 'ERM is not an imminent need for the company; it is the cyber incident that should be handled now, and such a one-off incident should not lead us to a big-bang reform like ERM.' Nevertheless, the CEO has asked for a more comprehensive need analysis of ERM within ten days.

- (i)** Outline points for response to the CEO citing the importance of ERM and comparing it with conventional risk management. (7)
- (ii)** Describe the key risks to which the company is exposed, pre- and post-integration with the acquired start-up. (10)

Following the cyber incident, the CEO has decided to employ a Head of Information Security who will report to the IT Head. The CEO has also asked you for further recommendations for handling such cyber events.

- (iii)**
 - (a)** Analyse this step taken by the CEO because of the risk governance (comments around reporting structure are required). (2)
 - (b)** List the recommendations sought by the CEO for proactively handling cyber-attacks. (2)

Given the anticipated high volatility in the medium term, one of the company's investment portfolios—which supports customers for approximately one year, is exposed to significant credit and market risks. To safeguard against potential adverse outcomes, it is essential to establish precise risk limits and implement robust controls for these risks.

- (iv)**

- (a) What methods can be employed to measure credit and market risks within the company's investment portfolio, especially in anticipation of high volatility in the medium term? (4)
- (b) How can adequate limits be set to manage the portfolio's identified credit and market risks? (3)
- (c) If these established risk limits are breached, what steps can be taken to address and resolve the situation effectively? (4)
- (v) Who are the key stakeholders in the company's Enterprise Risk Management (ERM) implementation, and what are their respective perspectives and interests in this process? (7)

[39 Marks]

Q. 2) You are the Chief Actuary working for an insurance company. The insurer has decided to implement internal capital modelling. This exercise aims to internally review and present the capital adequacy level and optimum capital utilization to the Board. You are considering various options for risk measures and methods of capital allocation. Assume that currently, there is no regulatory or internal prescription of principles on the measurement of risk or the confidence interval level.

- (i)
 - (a) Define risk measures. (1)
 - (b) State on what basis the time horizon of a risk measure is decided. (2)
 - (c) Define Value at Risk (VaR) and Tail-Var (TVaR). (2)
 - (d) Discuss the advantages and disadvantages of Tail-Var. (2)
- (ii) The insurer is considering investing INR 1,50,000 each in two assets in a portfolio. 1-year volatility for each asset is 2%. The correlation coefficient between these two is 0.37. Calculate a 5-year 97% VaR for the portfolio. State assumption(s) made and any related limitation of the same, if any. (5)

The insurer has significant long-term liabilities and has backed these obligations with a portfolio of corporate bonds. However, the insurer is re-evaluating its risk management strategies due to rising concerns over increased credit default risks, particularly amid economic uncertainty and market volatility. The insurer is considering implementing an internal capital model to assess and manage credit default risk within its bond portfolio.

- (iii) What are the key reasons for developing an internal credit default risk model to manage the insurer's exposure to potential defaults within its corporate bond portfolio? (6)

You are working on capital allocation on two lines of businesses (LoBs) – A and B in a portfolio. These independent LoBs have the following loss scenario:

- A: 20% chance of INR 99 Cr losses, zero otherwise
- B: 5% chances of INR 100 Cr losses, zero otherwise

- (iv)
 - (a) Calculate the Value at Risk (VaR) at a 99% confidence level. (6)

- (b) State various considerations in capital allocation methods for these LoBs. (9)
[33 Marks]

- Q. 3)** A large supply chain company has been prospering and growing with its cutting-edge practices. The company initially thrived by shifting its focus from supplying raw materials and resources to facilitating transactions between producers and consumers. Moreover, creating a groundbreaking B2B supply-chain management platform significantly strengthened its revenue streams.

In a rush to scale up operations due to increased competition in the market, key personnel of the company resorted to unfair financial transactions and practices, failing to maintain transparency in records and information to mask declining profits.

Two years after a reputable audit firm failed to detect these irregularities, regulatory authorities initiated an inquiry into the company's operations, ultimately exposing widespread misconduct. The key personnel involved and several others faced allegations of fraud and conspiracy.

The company's approach was near bankruptcy, and business operations were halted, as ordered by the regulatory authority. This had far-reaching effects, particularly devastating the retirement savings of employees and investors alike.

- (i) How could the company have effectively identified and quantified the operational risks of key personnel engaging in unfair financial practices? (7)
- (ii) What measures could the company implement to mitigate or transfer the operational risks associated with key personnel engaging in unfair financial practices? (11)

The company faces a difficult financial situation now, and the future of employees' retirement savings has caused widespread concern and uncertainty among the workforce due to the company's near-bankruptcy. Many company employees contributed to their pension plans expecting a secure retirement, but they fear reduced or delayed pension benefits.

Moreover, the company faces the risk of a reputation as an employer among job seekers, which might affect the interest of the new workforce. This further complicates stakeholders' confidence in the company and its sustainability.

- (iii) What actions should the company take to protect pension funds, reassure employees about their financial security, and rebuild trust with potential job seekers? (10)
[28 Marks]
